Screen Orientations and Attention: A Gamified Study on Mobile User Focus

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Abstract

This research investigates how screen orientation (portrait vs. landscape) influences mobile users' attention, performance, and overall user experience within a gamified context. We designed a controlled mobile game with minimal UI variations to isolate orientation effects and conducted a user study collecting objective data such as survival time, collision count, and score alongside subjective measures including cognitive workload (NASA-TLX) and custom questionnaires assessing perceived attention and satisfaction. Results provide insights beneficial for designing attention-demanding mobile interfaces and enhancing user engagement.

Author Keywords

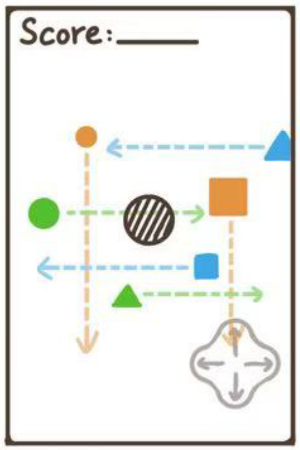
Screen orientation; User attention; Mobile games; User experience; Gamification.

CCS Concepts

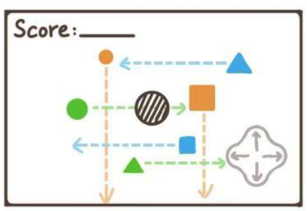
•**Human-centered computing** → **Human computer interaction (HCI);** *Haptic devices;* User studies; Please use the 2012 Classifiers and see this link to embed them in the text: https://dl.acm.org/ccs/ccs_flat.cfm

INTRODUCTION

Smartphones support two primary screen orientations: portrait and landscape. Mobile applications, especially games



(a) Portrait mode



(b) Landscape mode

Figure 1: Concept Figure of Our Game.

and task-oriented tools, often demand user attention to achieve optimal performance[11]. Previous studies have explored related dimensions of mobile device interaction, including the interactive effects of screen and product orientation on perceived product size[12] as well as the impact of screen size, viewing angle, and players' immersion tendencies on the overall game experience[6]. However, there remains a notable gap in understanding how different screen orientations influence user attention specifically. This gap limits informed user interface(UI) or user experience(UX) design decisions and could negatively impact user performance in attention-demanding mobile scenarios.

Our research aims to investigate the impact of screen orientation on mobile users' attention. We designed a controlled game with both portrait and landscape modes, minimizing UI-related confounding factors, to evaluate user attention, performance and overall experience.

RELATED WORK

A recent study, regarding the impact of screen orientation on user experience in the context of education-based mobile learning[14], found that both portrait and landscape modes are acceptable, but users reported a significantly better experience with portrait orientation. Although the study focused on educational applications, our research in a gamified context draws inspiration from its findings, aiming to further explore how screen orientation influences user attention during interactive mobile tasks.

Attention is a fundamental cognitive function, and its assessment has traditionally relied on standardized tasks that, despite their validity, present several limitations such us remarkable complexity, high resource demands, and the need for trained administrators[9]. Recognizing these limitations, researchers have increasingly integrated game elements into cognitive assessment tasks[15], a practice known as gamification[3]. Many studies have successfully utilized gamified tasks for attention assessment[1, 2].

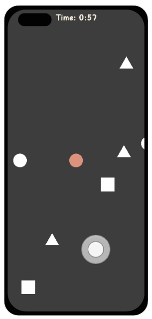
METHODS

To evaluate user attention, performance, and experience across portrait and landscape screen orientations, we designed a prototype mobile game and conducted a user study. Participants completed gameplay sessions in both orientations. The study involved collecting both objective and subjective data. We analyzed the data using statistical methods. The results were visualized and reported to support comparisons between the two conditions and to identify potential differences in user engagement and cognitive load.

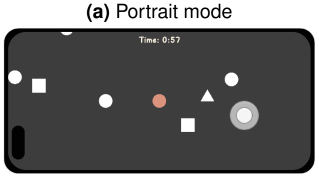
Game Design

In order to evaluate the effects of screen orientation on the experience of mobile games, our user study uses a controlled game design that features identical user interface (UI) layouts in both portrait and landscape modes. By maintaining visual and functional consistency, we can isolate the effects of orientation on cognitive workload and interaction patterns, which enable us to directly address the specific effects we want to evaluate.

As shown in Figure2, the brown circle represents the player that is navigated via fixed D-Pad. The colored arrows indicate obstacles approaching from different directions, while the black striped circle denotes the player. To eliminate confounding variables, screen orientation is set as the only independent variable. Obstacles are randomly launched toward the player from all four sides, either horizontally or vertically. Although the obstacles are launched randomly, we set a fixed random seed to ensure the same obstacle pattern is presented in every test session.



(a) Portrait mode



(b) Landscape mode

Figure 2: Screenshots of Our Game.

Measurements
Direct measurement of user attention ideally requires eye-tracking equipment[7]. However, due to hardware limitations, we used the number of times the user was hit by obstacles within 60 seconds as an indirect measure of user attention.

Additionally, we used NASA Task Load Index(NASA-TLX)[4, 5] to measure perceived cognitive workload and a custom 7-point Likert-scale[8, 13] questionnaire to subjectively evaluate user attention, performance and satisfaction across different screen orientations.

Participants
16 users participated in our user experiment. Participants were from the general student population.

Procedure
After a brief introduction of the study, participants were asked to play two rounds of our obstacle avoidance game using the same smartphone model. They were free to hold the device in any posture they preferred for each round. The experimenters observed and categorized each participant's device-holding posture during gameplay, and recorded the posture used in both rounds. At the end of each round, objective performance data displayed by the system were manually recorded. Participants then were asked to complete a questionnaire consisting of the standard NASA-TLX workload items and additional custom 7-point Likert-scale questions to provide subjective evaluations. Each participant experienced both conditions in a counterbalanced order to avoid sequence effects[10].

RESULTS

To evaluate how people perform differently in landscape and portrait modes of our game, we designed a user test. In each test, participants were informed of our research goals and how to play the game. Participants completed the landscape and portrait modes in a different order. After completing one mode, they were asked to complete the same questionnaire.

Posture during testing
During the test, 10 (62.5%) testers used two hands to participate in the game, that is, holding the phone with two hands and controlling the player character with the thumb, or holding the phone with one left or right hand and controlling the player character with the other hand. The other 6 (37.5%) testers only used one hand to hold the phone and control the player character. For controlling the player character, 14 (87.5%) testers of the players used thumbs, 1 (6.25%) testers were more accustomed to using the index finger to control, and 1 (6.25%) testers of the players used the thumb and index finger alternately to control the player character.

Collision number analysis
We counted the number of times 16 testers encountered obstacles in landscape and portrait modes. Figure 3 shows the test results, and Figure 4 shows a scatter plot of the test results. Each column in the table represents the performance of a tester in two modes. In this test, the odd-numbered testers first experienced the portrait mode and then the landscape mode, while the even-numbered testers did the opposite.

- *Comparison of collision times in different modes*
  Among the 16 testers, except for 2 (13%) testers who performed equally in both modes, 9 (56.25%) testers experienced fewer collisions in landscape mode, meaning they performed better in landscape mode than in portrait mode. Besides, the remaining 5 (31.25%) testers performed better in portrait mode.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Portrait mode	21	30	8	11	27	9	20	12	11	14	14	3	11	8	11	20
Landscape mode	3	30	9	21	26	9	16	16	8	10	10	14	6	7	16	17

Figure 3: Table of the number of times 16 testers encountered obstacles in landscape and portrait modes

- *Effect of different order*
  After comparing all test takers' collision times in the latter mode with those in the former mode, we found that except for 2 (13%) testers who performed equally in both modes, 9 (56.25%) testers demonstrated better performance in the latter mode by reduced collision counts, while 5 (31.25%) testers performed better in the former mode.

NASA-TLX Analysis

In this test, we used NASA-TLX to evaluate the load of the game in landscape mode and portrait mode on the testers. We set a question in each of the six different dimensions to evaluate the task load, and each question is scored from 1 to 10, from low to high. Figure 5 and Figure 6 show the testers' feelings in portrait mode and landscape mode respectively.

In the three dimensions of Mental Demand, Own Performance, and Effort, the mean scores in landscape mode are slightly higher than those in portrait mode. In the other three dimensions of Physical Demand, Temporal Demand, and Frustration Level, the mean scores in landscape mode are slightly lower than those in portrait mode. The total task load is calculated based on the means of the six dimensions in both modes. The overall TLX score in landscape mode (5.177) is lower than that in portrait mode (5.333).

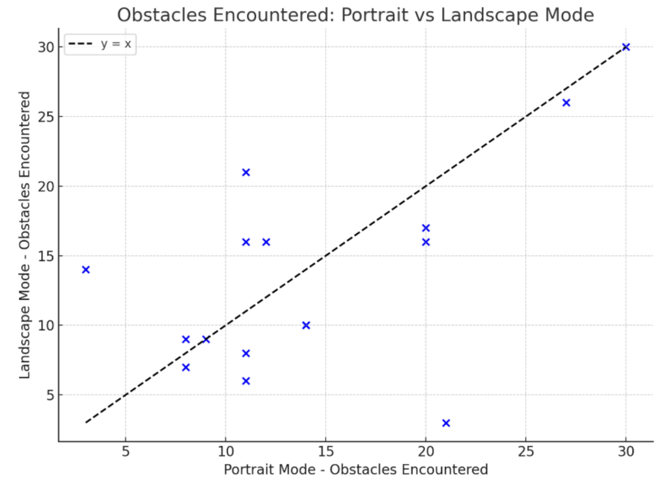


Figure 4: Scatter plot of test results: Points below the black dashed line represent more collisions in portrait mode than in landscape mode

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Mean
Mental Demand	3	8	3	7	5	4	5	3	5	7	7	7	7	9	3	3	5.375
Physical Demand	2	6	1	6	5	4	4	2	6	2	4	6	5	7	2	4	4.125
Temporal Demand	5	7	8	7	8	4	7	7	8	6	7	5	7	8	6	2	6.375
Own Performance	3	3	7	4	4	6	8	4	6	6	8	9	8	5	6	5	5.75
Effort	3	7	3	6	7	4	6	4	5	7	6	7	5	8	3	4	5.063
Frustration Level	9	6	2	8	6	5	4	5	7	3	6	3	6	9	4	2	5.313

Figure 5: Table of NASA-TLX data for 16 players in portrait mode

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Mean
Mental Demand	7	7	4	7	5	5	4	2	3	7	7	9	7	8	3	5	5.625
Physical Demand	3	5	1	2	5	6	6	1	3	2	4	6	5	4	2	2	3.563
Temporal Demand	3	8	4	7	5	5	4	8	4	6	7	4	7	8	4	2	5.375
Own Performance	9	4	4	4	6	4	8	3	7	6	8	8	9	8	4	6	6.125
Effort	7	7	3	6	6	4	4	6	6	7	4	8	7	5	4	4	5.5
Frustration Level	6	7	3	8	5	6	2	3	8	4	3	4	6	6	5	2	4.875

Figure 6: Table of NASA-TLX data for 16 players in landscape mode

Discussion

During the test, most people used their right thumb to move the player character. Therefore, in the design of touch-screen control games in mobile interaction, placing the joystick in the appropriate lower right corner is better for the player experience. After players experience one mode, they will perform better in the other mode. We speculate that this is because players will gain some game experience in the first mode, which makes them perform better in the latter mode. And regardless of the order in which players experience the two modes, players in landscape mode are more likely to perform better.

And for the NASA-TLX analysis, in terms of Physical Demand, the landscape mode (3.563) is lower than the portrait mode (4.125), which may be because the landscape mode is more comfortable and easier to operate. In terms of Mental Demand, the result of landscape mode (5.625) is slightly higher than that of portrait mode (5.375), probably because landscape mode requires more visual scanning. At the same time, the wider field of view of landscape mode may reduce time pressure, making the result of landscape mode on Temporal Demand (5.375) better than that of portrait mode (6.375). In terms of Own Performance, the result of landscape mode (6.125) is better than that of portrait mode (5.75), indicating that players perform well in landscape mode and are satisfied with it. Therefore, the frustration in landscape mode is reduced. The overall load score for landscape mode (5.177) is lower than that for portrait mode (5.333), indicating that landscape mode has slightly lower task load and players are more likely to achieve better results, which is consistent with the obstacle collision results.

For better performance in landscape mode, we think that this is because people's attention in the vertical direction is better than in the horizontal direction; that is, compared with processing information in the horizontal direction, people are better at processing vertical information in the vertical direction. Although the vertical distance in the landscape mode is shorter, people's attention in this direction is more concentrated, so it is easy to avoid these fast-moving obstacles from the vertical direction in the game. However, in the portrait mode, the horizontal distance of the screen is shorter, people's attention in the horizontal direction is poor, and it is not easy to avoid fast-moving obstacles from the horizontal direction. Therefore, testers can avoid more collisions and perform better in the landscape mode.

CONCLUSION

Our study demonstrates that landscape orientation enhances user attention, performance, and experience in mobile gamified tasks compared to portrait mode, proven by lower collision times, reduced cognitive workload (TLX score 5.177 vs. 5.333), and improved satisfaction. The wider field of view in landscape mode likely reduces time pressure and frustration, while better vertical attention processing assists obstacle avoidance. These findings underscore the importance of screen orientation in designing attention-demanding mobile interfaces. Future research could incorporate eye-tracking to further validate these results, ensuring robust UI/UX design for mobile gaming.

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